

Computer Networks

Lecture 22 Network Layer

Content



- **Subnet**
- **IP Address**

Question

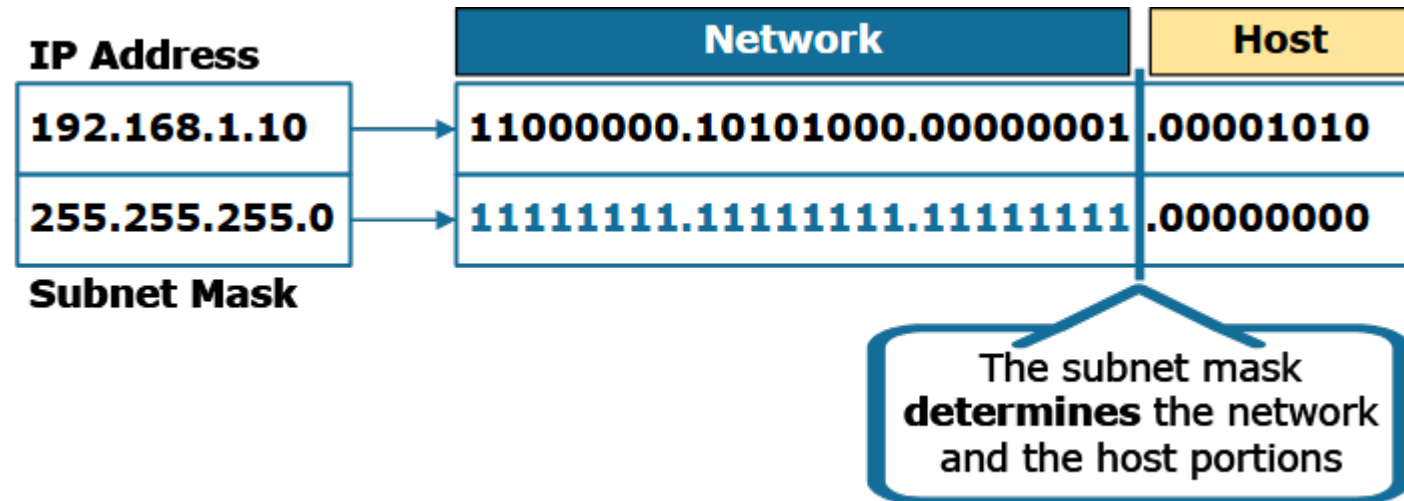
1. A Class B network needs 500 hosts per subnet. Find:
 - Number of subnet bits required
 - Number of possible subnets
 - Subnet mask

2. A company receives Class B network 140.12.0.0 and needs 120 subnets. Find:
 - Number of bits borrowed
 - Subnet mask
 - Hosts per subnet

Subnet Mask

The subnet mask identifies which bits in an IP address represent the network ID and which bits represent the host ID.

A subnet mask contains a contiguous sequence of 1's followed by a contiguous sequence of 0's. The sequence of 1's represents the Network ID part, and the sequence of 0's represents the Host ID part.



Fixed Length Subnetting

Fixed length subnetting divides the network into subnets where

- All the subnets are of same size.
- All the subnets have equal number of hosts.
- All the subnets have same subnet mask.

In fixed length subnetting, all the subnets have same subnet mask since the size of each subnet is same.

Variable Length Subnetting

Variable Length Subnetting, also known as Classless Subnetting, divides a network into subnets that can vary in size.

In this approach:

- Subnets may have different numbers of hosts,
- Subnets can be of unequal sizes, and
- Each subnet may use a different subnet mask.

Subnetting

When a data packet arrives at the internal router, the router determines the appropriate outgoing interface using the following process:

Router performs a bitwise AND between:

- ✓ The destination IP address from the data packet
- ✓ Each subnet mask in its routing table, one by one

Router compares each result with the corresponding destination subnet IP address in the routing table.

Based on the comparison, three possible cases may occur:

- ✓ Case 1: Only One match
- ✓ Case 2: Multiple matches are found

Network	Mask
192.168.1.0	255.255.255.0
192.168.1.0	255.255.255.128

- ✓ Case 3: No match is found

Questions

1. Consider a large network with address 200.1.2.0. The objective is to perform subnetting to partition this network into four subnets. What will be the subnet mask.
2. The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network?
 - a) 10.35.28.2 and 10.35.29.4
 - b) 172.57.88.62 and 172.56.87.23.2
 - c) 191.203.31.87 and 191.234.31.88
 - d) 128.8.129.43 and 128.8.161.55

Questions

1. Consider a network is subnetted into 4 subnets with following subnet id

200.1.2.0 (Subnet A) 200.1.2.64 (Subnet B) 200.1.2.128 (Subnet C) 200.1.2.192 (Subnet D)

Then, the routing table maintained by the internal router looks like:

Destination	Subnet Mask	Interface
200.1.2.0	255.255.255.192	a
200.1.2.64	255.255.255.192	b
200.1.2.128	255.255.255.192	c
200.1.2.192	255.255.255.192	d
Default	0.0.0.0	e

A packet bearing a destination address 200.1.2.130 arrives at the router. On which interface will it be forwarded?

Categories of IP addresses

Public IP address

- Public IP addresses are globally accessible through the internet. They are assigned to devices that need to be reachable from anywhere in the world, such as web servers, email servers, and other internet-facing services.
- For instance, when you use a command like `nslookup www.google.com` in the terminal, it reveals the public IP address of Google's server.

Private IP address

- Private IP addresses are used within local networks and cannot be directly accessed from the internet.
- Devices like computers, printers, and mobile devices within home, office, or university networks typically use private IP addresses.

Categories of IP addresses

Static IP address

- A static IP address is a permanent IP address assigned to a network device. It remains unchanged unless manually reconfigured by a network administrator.
- Typically, public-facing infrastructure like routers, web servers, and email servers use static IP addresses to maintain consistent network identification and accessibility

Dynamic IP address

- Dynamic IP addresses are temporarily assigned to network devices by a DHCP (Dynamic Host Configuration Protocol) server. These addresses can change each time a device connects to the network.
- For example, when you connect to a university network, you might receive a different IP address during each session, depending on network configuration and address availability.

Casting in Computer Network

Casting is the method of delivering data from a source to one or multiple destinations on a network. It specifies how a device communicates with others

Casting is mainly divided into four types:

- A single device (unicast)
- All devices on the local network (limited broadcast)
- All devices on a specific network (directed broadcast)
- A particular group of devices (multicast)

Unicast

- Unicast refers to transmitting data from one device to a specific device, either within the local network or on a different network.
- It is a one-to-one transmission method.

Example

Example: Host-1 with IP address 25.4.7.8 transmits data to Host-5 with IP address 28.14.8.10, where the source IP address is 25.4.7.8 and the destination IP address is 28.14.8.10

Broadcast

- Transmitting data from one device to all devices simultaneously within the local network or on a different network is called broadcasting.
- It is a one-to-many transmission method.

Based on the recipient's network, it is further classified into two types.

1. Limited Broadcast
2. Directed Broadcast

Limited Broadcast

- Transmitting data from one device to all other devices simultaneously within the same network is called limited broadcast.
- Limited broadcast address for any IP address is network ID and host ID bits are set to 1.

Example

Host-1 with IP address 28.52.46.81 broadcasts data to all other devices simultaneously within its local network, where source IP address is 28.52.46.81 and destination IP address is 28.255.255.255

Directed Broadcast

- Transmitting data from one device to all the devices simultaneously on a different network is called directed broadcast.
- Directed broadcast address for any network is Host ID bits are set to 1.

Example

Host-1 with IP address 28.52.46.81 broadcasts data to all devices on a different network with network address 38.0.0.0, where source IP address is 28.52.46.81 and destination IP address is 38.255.255.255

Multicast

- Transmitting data from one device to a particular group of devices within the same or a different network is called multicast. (Like speaking to a selected group who have chosen to listen or receive the information)
- Only the devices in the multicast group will receive the data.

Example

video streaming or online conferencing applications often use multicast to efficiently deliver the same video or audio stream to multiple participants.